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Synergistic optimization of baffles and aeration to improve the Light/Dark cycle of microalgae photobioreactor for enhanced nitrogen removal performance: Computational fluid dynamics and experimental verification

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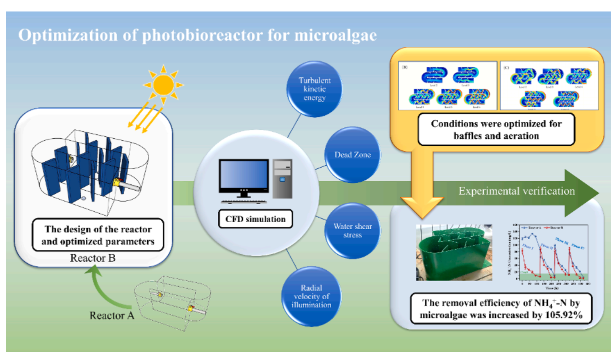
HIGHLIGHTS

- The co-optimization of baffle and aeration was achieved.
- The processing efficiency of the optimized reactor increased by ca. 106 %.
- The novel runway pool photobioreactor reduced the Light/Dark cycle time by 89.6 %.

GRAPHICAL ABSTRACT



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ABSTRACT

Microalgae photobioreactor (PBR) is a kind of efficient wastewater treatment system for nitrogen removal. However, there is still an urgent need for process optimization of PBR. Especially, the synergistic effect and optimization of light and flow state poses a challenge. In this study, the computational fluid dynamics is employed for simulating the optimization of the number and length of the internal baffles, as well as the aeration rate of PBR, which in turn leads to the optimal growth of microalgae and efficient nitrogen removal. After optimization, the Light/Dark cycle of the reactor B is shortened by 51.6 %, and the biomass increases from 0.06 g/L to 3.94 g/L. In addition, the removal rate of NH_4^+-N increased by 106.0 % to 1.56 mg L^{-1} h^{-1}. This work

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provides a feasible method for optimizing the design and operational parameters of PBR aiming the engineering application.

1. Introduction

Microalgae have the ability to adapt to extremely low C/N wastewater. Geng et al. (2022) successfully used carbonate-enhanced microalgae to treat rare earth element tailings (REEs) wastewater. However, most scholars only pay attention to the removal mechanism and ignore the influence of the reactor on the treatment effect. The runway pool is a kind of open, simple Microalgae photobioreactor (PBR) that has been used for the cultivation of commercial microalgae (Long et al., 2022). PBR is an indispensable equipment for the engineering application of wastewater treatment technology using microalgae (Huang et al., 2015). However, the operation of microalgae in PBR is a complicated process in which light and fluid dynamics both have important effect on wastewater treatment, especially for nitrogen assimilation (Patil et al., 2020). The growth rate of microalgae is closely related to the light intensity of absorption, and the light available to microalgae will change because of the penetration depth and the shading effect of the cells (Azizi et al., 2018). Fluid dynamic parameters play an important role in analyzing the degree of liquid mixing (Kusmayadi et al., 2020). The adequate mixing of microalgae and wastewater facilitates the uniform suspension of cells (Orefice et al., 2019), helping nutrients to be absorbed and converted more quickly. Mixing along the radial direction of light also promotes photosynthetic efficiency (Cui et al., 2020). The fluid dynamics in the PBR are required to enable microalgae cells to move back and forth between the light and dark regions to achieve maximum photosynthetic efficiency (Del Olmo et al., 2022; Richmond, 2004).

Taking different shapes such as flat, cylindrical or raceway pools can affect microalgae growth, and changes in size of reactor can also have an effect on changes in flow patterns (Guo et al., 2015). One of the most important factors affecting the degree of mixing of liquids is the design of the reactor, where the amount of air in the wastewater is increased by different configurations, generating bubbles that allow the liquid to be quickly and evenly distributed inside the reactor (Rezvani and Rostam, 2023). Addition of baffles and aeration are effective means of improving the reactor's performance (Yaqoubnejad et al., 2021). The baffles enable the formation of vortices in the water column inside the reactor, suspending microalgae cells, promoting their vertical movement and enhancing CO_2 fixation and nutrient uptake (Kubar et al., 2022). Yang et al., (2016) found that the baffles installed in the reactor resulted in a 52 % increase in microalgae cell Light/Dark (L/D) cycle efficiency and a 22 % increase in biomass. Aeration is usually applied to microalgae culture to reduce dead space, remove O_2 produced by photosynthesis and help in fluid circulation (Orefice et al., 2019). Cui et al., (2020) optimized the aeration direction and aeration rate amount of a tubular PBR and found that the L/D cycle frequency and light time increased by 78.2 % and 36.2 % when the aeration rate was set at 0.7 vvm. Although reactor optimization has been explored previously, the study of baffle and aeration synergistic optimization of microalgae PBR is relatively rare.

The unoptimized PBR exhibits the drawbacks of internal dead zones and suboptimal photosynthesis (Ranganathan et al., 2022). Traditional methods for reactor optimization often rely on experiential knowledge and extensive experimentation to compare performance changes before and after parameter adjustments, aiming to enhance its overall efficiency (D'Bastiani et al., 2023). Due to the lack of system framework, this method is inefficient and consumes a lot of time and money (Hong et al., 2022; Barceló-Villalobos et al., 2019). Computational fluid dynamics (CFD) is a numerical method that solves fluid flow and heat transfer as well as other relevant physical and biochemical processes (Pan et al., 2017). To date, many scholars have used CFD to optimize bioreactor to achieve higher culture efficiency or processing speed (Sánchez et al., 2018). Using wastewater to cultivate microalgae is a good technology for energy saving and environmental protection (Han et al., 2022). The studies on photobioreactor with baffle and aeration only focus on the mass transfer of gas and lack of analysis of light, but for microalgae, light is a key factor (Li et al., 2023; Zhang et al., 2024). An excellent PBR can maintain the maximum growth rate of microalgae. Due to the dead angle of liquid flow in the reactor, and the need for radial movement of light to enhance photosynthesis in microalgae cells, how to control these factors synergically is undoubtedly a challenge.

In this work, a newly runway pool PBR was designed to promote the mixing and mass transfer between microalgae and wastewater by adding baffles and aeration to reduce dead zones and improve the performance. The effects of baffle setting and aeration rate volume adjustment on wastewater flow pattern was analyzed by CFD simulation. The motion trajectory of microalgae cells in the reactor and the light condition was studied. The biomass growth of microalgae before and after optimization was compared and the simulation results with experimental data was verified.

2. Materials and methods

2.1. Rare earth element tailings wastewater

The wastewater was collected from the Longnan Foot Cave rare earth element mine area in Longnan County, Ganzhou City, Jiangxi Province, China (N24°85′, E114°86′). The constituents in REEs wastewater include NH_4^+-N 105.6 mg/L, NO_3^--N 64.5 mg/L, NO_2^--N<0.1 mg/L, P<0.1 mg/L, TOC 30 mg/L and IC 35 mg/L. The pH value was 8–10.

2.2. Microalgae

The microalgae *Chlorococcum robustum* AY122332.1 with high ammonia tolerance in REEs wastewater was used in this study (Geng et al., 2022; Zhang et al., 2019). The microalgae were cultured for 15 days in Blue Green medium (Shanghai Guangyu Biological Technology Co., Ltd) at 28 ± 1 °C with an illumination intensity of 100 μmol photons $m^{-2}\cdot s^{-1}$ prior to the experiment.

2.3. Reactor configuration and operation

Two lab-scale reactors were customized, namely the original reactor (RA) and the optimized reactor (RB), separately. As shown in Fig. 1, runway pool reactor A consists of P_1 inlet, P_2 outlet and a main body composed of three parallel cuboids and two semicircles. The overall length of the reactor is 400 mm, width is 300 mm, height is 200 mm, and the working liquid volume is 15 L. RB added gas spargers and baffles to RA, in which the number and size of baffles were determined by the CFD simulation. Each baffle is perpendicular to the bottom of the reactor and has an angle of 60° between it and the reactor separator. Furthermore, in this study, light was assumed to be directed vertically above the reactor. The size and operating conditions of the two reactors were identical except for the baffles and gas spargers used in RB.

Two parameters were chosen for the optimization of baffles in this study, namely the number and length of baffles, and 6 gradients were designed for each parameter (Table 1). Six gradients were chosen for the optimization of aeration rates (Table 1). After analyzing the optimized data, a new type of PBR, i.e., RB was obtained by optimizing the design of reactor A. Finally, reactors A and B were used for under the same experimental conditions.

2.4. Simulation condition

The CFD models in this study were constructed and run using COMSOL Multiphysics 6.0. $k-\varepsilon$ turbulence model was used to characterize the turbulent effect of wastewater in the reactor when calculating the optimization stage of baffle. Since the density of the suspended algal fluid was similar to that of water, water was used instead of algal fluid in the calculations. The inlet velocity of the water flow was set to 0.5 m/s.

When the aeration equipment was optimized, the simplified Euler-Euler biphasic flow model was selected for calculation because of the presence of bubbles in the PBR. Because the microalgae system is small, the microalgae and wastewater were combined into a continuous phase. The liquid phase was the continuous phase, and the gas phase was the dispersed phase. Grid independence argument is shown in Table 2. Considering the computational time and model convergence, the reactor solution freedom is 250,368 (including 652,318 internal degrees of freedom) that converged at a residual of 10^{-3} (see Supplementary Material). The airflow velocity was determined by the aeration rate. The upper boundary of the reactor was set as the dispersed phase gas outlet, so that the liquid phase cannot pass through. The gas phase outlet was the entire upper end of the reactor and the outlet pressure was set to atmospheric pressure. In addition, the other wall conditions were designed to be no slip and the liquid phase exit to be static pressure. The computational time step was 15 s and was recorded at 0.01 s intervals.

In order to observe the movement of microalgae cells in the reactor, a particle tracking model was used for simulation. Assuming that the microalgae were suspended in the reactor, 50 microalgae particles were randomly released in the reactor at time 0 s, with a density of 997 kg/m^3 and a size of 5 μm, and the release position was discrete distribution. The water flow velocity was coupled to the previously calculated velocity field. Calculations were recorded at 0.01 s intervals for a total of 15 s. After the calculation, the particle trajectory can be known through the position parameters in the illumination direction.

The verified experiment were operated using RA and RB in parallel at room temperature (28 ± 2 °C), 20,000 Lux illumination, microalgae amount of 2 g dry weight/L for 30 d.

2.5. Analytical and calculation method

System effluent samples were filtered through 0.22 μm membrane filters, and the filtrates were preserved at 4 °C until analysis. The concentration of NH_4^+-N, NO_3^--N and NO_2^--N was measured using a UV spectrophotometer (UV–2600, Unocal Instrument Co., Shanghai) according to standard methods (SEPA, 2002).

The biomass of microalgae in the reactor was calculated as follow (Nayak et al., 2016):

$$p = \frac{C_1 - C_0}{t_1 - t_0} \tag{1}$$

where cell density (g·L^{-1}) at t_1 (d) and t_0 (d), respectively.

The *Lambert-Beer* light attenuation model was used to calculate the light reaching the microalgae in the reactor (Yun et al., 2001). It is assumed that light is incident vertically at the water surface, so the reactor can be defined as a simplified calculation of the section containing only two axes Y-Z (see Supplementary Material). The formula is as follows:

$$A(X) = \frac{1}{L}\ln\frac{I(X)}{I_0} \tag{2}$$

$$A(X) = K_a(X + b) \tag{3}$$

$$I(X) = I_0\hat{A}\cdot e^{-k_a(X+b)} \tag{4}$$

where, k_a represents the optical absorption coefficient, set at 0.4 cm^{-1} in this study. I_0 is incident light intensity (Lux), X is microbial concentration (kg m^{-3}), b is attenuation coefficient of water, and L is optical path (m).

The region in the photobioreactor where the light intensity was stronger than the saturation light intensity of the microalgae cells was defined as the light zone, and vice versa was the dark zone (Levasseur et al., 2018). In this study, the area of reactor z > 175 mm was defined as the light zone by the *Lambert-Beer* formula.

The motion trajectory of microalgae cells in the reactor can be obtained by particle tracking model. The L/D cycle is calculated by recording the number of cycles of the particle in the light and dark regions (Chen et al., 2016).

Table 1
Optimized parameters of reactor design.

Optimization parameter	Level					
	1	2	3	4	5	6
Length of baffle (cm)	\	3	4	6	7	8
Number of baffles	\	10	13	14	17	20
Aeration rate (vvm)	0.2	0.4	0.6	0.7	0.8	1.0

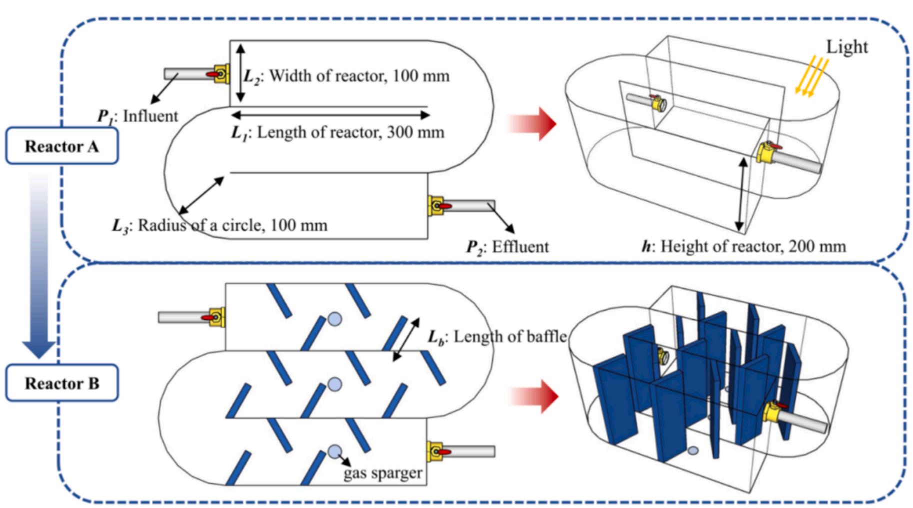

Fig. 1. Schematic diagram of the reactor design.

Table 2
Grid independence argument.

	Data						
Grid degree of freedom	95,460	186,576	229,242	250,368	308,520	400,902	536,460
Average flow rate (m/s)	0.092	0.104	0.107	0.114	0.116	0.116	0.122
Error of relative (%)	12.5	2.6	6.6	2.1	0.03	4.5	\

3. Results and discussion

3.1. Visualization of the flow characteristics by simulation

The fluid flow state in the reactor was simulated by CFD with inlet velocity of 0.5 m/s and setting the parameters of algal liquid to be the same as water. Fig. 2A shown the velocity field and flow diagram of reactor A. In order to improve this situation, five gradients were simulated for the length of the baffle (Fig. 2B). As the length of the baffle increased, the fluid in the reactor gradually formed a number of small local circulations. The wastewater was divided into relatively independent chambers by the baffle. The local eddy currents formed in the chambers drive the liquid circulation, which facilitated mass transfer and light and dark movement of algal cells in the chamber. Then, the same method was used to optimize the number of baffles by five gradients (Fig. 2C). The more baffles, the more compartments would be

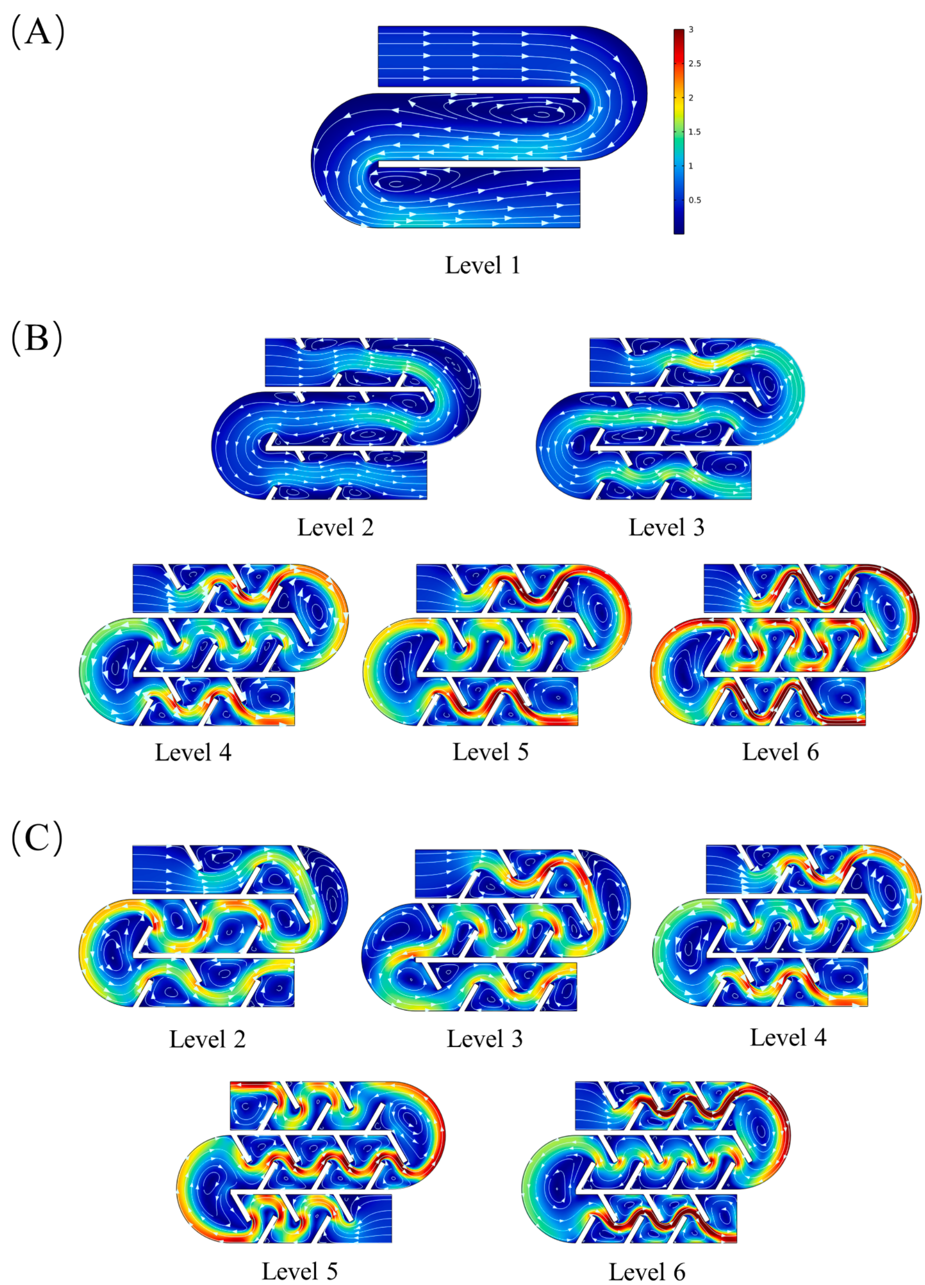

Fig. 2. The optimized reactor simulated velocity field and streamline diagram. (A) Velocity field and streamline diagram of reactor A; (B) Baffle length optimization; (C) Baffle quantity optimization.

segmented. It could also be seen from the YZ section (see Supplementary Material) that the eddy current formed in the reactor was suspended in the middle position, which was undoubtedly favorable for the light cycle of microalgae cells. The setting of the baffle increased the turbulent dissipation rate, which was a parameter that measured the energy conversion. By setting baffles, the turbulent dissipation rate of the baffle quantity group increased from 0.21 m^2/s^3 to 26.16 m^2/s^3, and the length group increased to 14.78 m^2/s^3. The quantity could affect the flow state of wastewater more than the length. However, as Level 5 shown, excessive segmentation resulted in the failure to form local loops, so it was important to select the right number of baffles.

3.2. Optimization of design and operational conditions

3.2.1. Baffle design

The number and position of baffles were given in Table 1 for CFD simulation, and the simulation resulted shown in Fig. 3. To gain a better understanding of the liquid mixing in the reactor, four flow regime parameters were calculated: turbulent kinetic energy (TKE), dead zone, average water shear stress (WSS), and radial velocity of illumination.

TKE was a numerical value that represents the strength of turbulence in a fluid, that was, the kinetic energy per unit mass associated with eddy currents in turbulence. The nutrient flow required by microalgae was enhanced by the increased degree of mass transfer, which promoted its absorption and microalgae growth. As the number and length of baffles increased, TKE shown an upward trend (Fig. 3A). TKE increased from 0.04 j/kg to 0.36 j/kg and 0.44 j/kg, with the visible length of the baffle having a greater impact on it. The higher the TKE value, the better the mixing degree of the liquid inside the reactor. The number and length of level > 4 were more consistent with this condition. Baffle plates were placed inside the reactor to assist in the formation of trapezoidal chambers. The number of eddies in each chamber depended on the number of baffle plates. The increase in the growth rate was related to the change in the streamline inside the reactor, and the local circulation formed by the baffle promotes the mixing of the liquid. (see Supplementary Material). The strong local liquid circulation also accelerated the photoperiod of microalgae. Dead zone referred to the zone in PBR where the flow rate was close to zero (Kawale et al., 2019). In this study, the dead zone was defined as the zone below 0.1 m/s, and the growth of microalgae inside the dead zone would be negatively affected. With the improvement of optimization level, the proportion of dead zones gradually decreased from 8.14 % to 1.29 % and 2.71 %, and it was worth noting that the lowest proportion of the length group occurred at level 4 (Fig. 3B). The increased dead area was caused by the corner of the baffle before the chamber was formed. WSS was the friction force between the fluid and the wall, and its magnitude depended on the flow rate and the friction coefficient of the wall. The WSS ranged from 0.04 Pa to 0.26 Pa and 0.29 Pa, and the effect of quantity and length changed on it was similar. Similar to the growth trend of TKE, the optimization level > 4 was better. Higher WSS reduced the adhesion of algae to the wall and facilitates the uniform distribution of algal cells in the wastewater. The installation of the baffle increased the contact between the wastewater and the reactor wall.

The radial velocity of illumination was the flow velocity along the direction of illumination. Due to the gradual attenuation of the effect of light under water, there were light and dark regions in the reactor. Water flow helped the microalgae cells to move between the light and dark zones, and the higher radial velocity improved the efficiency of cell movement to achieve higher photosynthetic efficiency. Fig. 3D shown that both the number and length of baffles were proportional to the radial velocity of illumination. However, in the optimization of baffle length, the speed had obvious fluctuations, so it was impossible to directly select the appropriate length from this point. It could be clearly seen from the streamlines in the reactor that the vortex formed by setting the baffle made the water flow have a tendency to circulate up and down, which was undoubtedly effective for cell movement (see Supplementary Material).

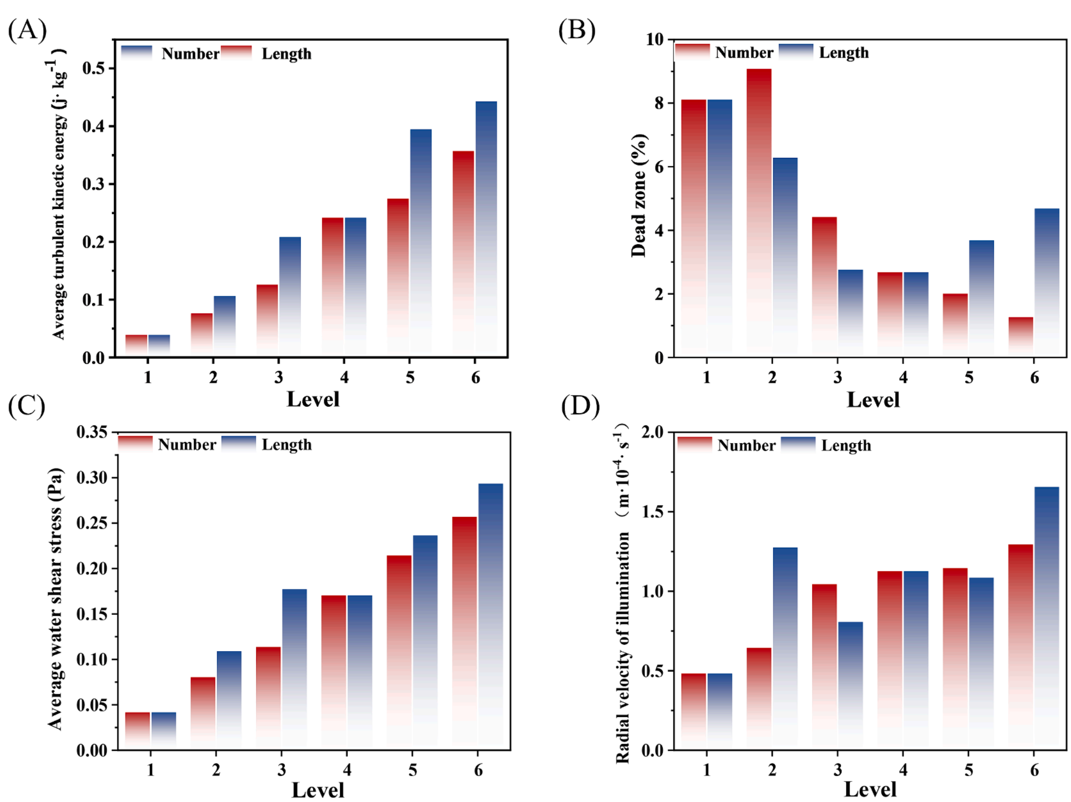

Fig. 3. The optimization of the length and number of baffles affects the flow state parameters. (A) Turbulence kinetic energy (TKE); (B) Dead zone; (C) Average water shear stress (WSS); (D) Radial velocity of illumination.

3.2.2. Aeration rate optimization

In order to further change the flow state to improve the illumination of microalgae, the aeration equipment was arranged at the bottom of the reactor. Aeration rates of six different gradients were set at 0.2, 0.4, 0.6, 0.7, 0.8 and 1.0 vvm, respectively (Fig. 4). It was optimized by comparing the changes of TKE, dead zone proportional, WSS and light radial velocity analysis of aeration convection.

With the increase of aeration rate from 0.4 vvm to 1 vvm (Fig. 4A), TKE increased from 0.04 j/kg to 0.08 j/kg, increasing by 70.0 %. The proportion of dead zones in the reactor reached a maximum at an aeration rate of 0.6 vvm and decreased between 0.6 and 1.0 vvm (Fig. 4B). The increase in the proportion of dead zones in the 0.2–0.6 vvm interval may be due to the local circulation area formed by the airflow in the reactor (see Supplementary Material), in which the microalgae cells were suspended. Such dead zones will not cause the clumps and growth inhibition of microalgae. WSS in the bubble flow can break the bubbles into smaller bubbles, and the increase of the surface area can bring CO_2 to the microalgae and make the cell distribution more uniform. Fig. 4C shown that with the increase of aeration rate, WSS also steadily increases. High WSS values could lead to cell damage, but the WSS did not reach the damage threshold in the reactors studied in this study (Michiel et al., 2010). Microalgae would first produce dispersed cells that could be suspended in the upper water body during the growth process; however, after a period of time, these cells would gradually form microalgae particles and settle due to gravity. Aeration could induce the formation of microalgae particles and promote the removal of nutrients from wastewater (Wang et al., 2022). The direction of bottom aeration was consistent with the radial direction of light, which helped the microalgae cells to undergo L/D cycle. The aeration rate was proportional to the velocity (Fig. 4D). In terms of TKE, WSS and radial velocity of light, higher aeration rate should be selected during reactor operation.

3.2.3. Light condition

In order to understand the light exposure of microalgae cells in the reactor, the particle tracking method was used to simulate the trajectory of cells in the reactor, focusing on the movement in the light direction to calculate the L/D cycle and light exposure time of each particle. The L/D cycle was defined as the time taken for a microalgae cell to complete a transition from the light to the dark zone, microalgae cells with L/D cycle grow faster and produce the so-called flash effect (Chiarini and Quadrio, 2021). It has been shown that the photosynthetic efficiency of microalgae under intermittent light is higher than that under continuous light (Qin et al., 2018). There were light-dependent chemical phases and biochemical dark phases of photosynthesis in microalgae (Fan et al., 2020). The time ratio of cells staying in the light zone to the dark zone could be used to illustrate the overall light exposure of microalgae.

Before optimization, the movement trajectory of microalgae in the reactor was single (Fig. 5A) and basically unchanged, which was not conducive to the growth of microalgae. In the optimized reactor (Fig. 5B), the microalgae cell movement trajectory had a very obvious up and down movement, which was undoubtedly a good change for its L/D cycle. Because the microalgae themselves had a certain degree of sedimentation, they will gradually accumulate to the bottom of the pool over time, which was not conducive to long-term development. The aeration changed the cell movement trajectory more obviously, while the vortex formed by the baffled plate pushes more cells to the upper light zone. It could be seen from Fig. 5C that the L/D cycle time was the shortest when the baffle length was 6 cm, which was 4.36 times shorter than that without baffle. When the number of baffles is 14, the best effect is achieved. Compared with the condition without baffles, the cycle time was shortened by 89.6 %. Significantly, with a length of 4 cm, the time

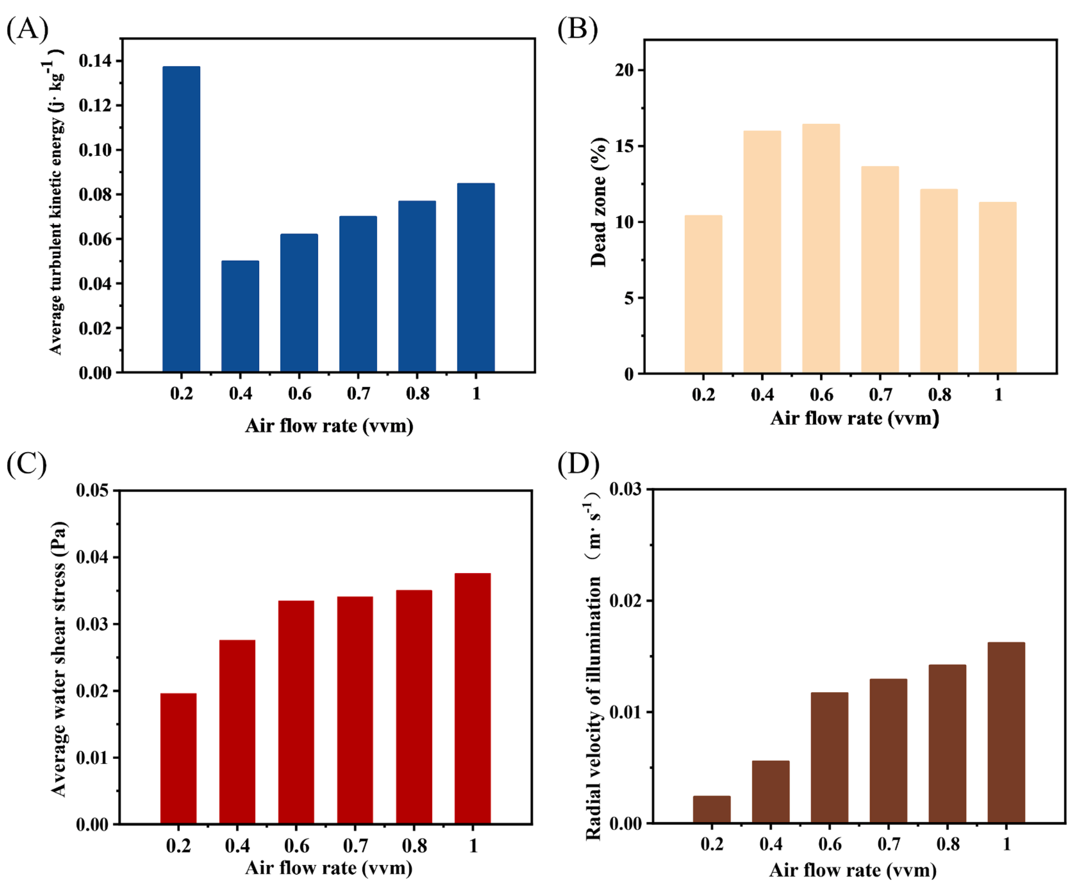

Fig. 4. Aeration rate optimization parameters. (A) Turbulence kinetic energy (TKE); (B) Dead zone; (C) Average water shear stress (WSS); (D) Radial velocity of illumination.

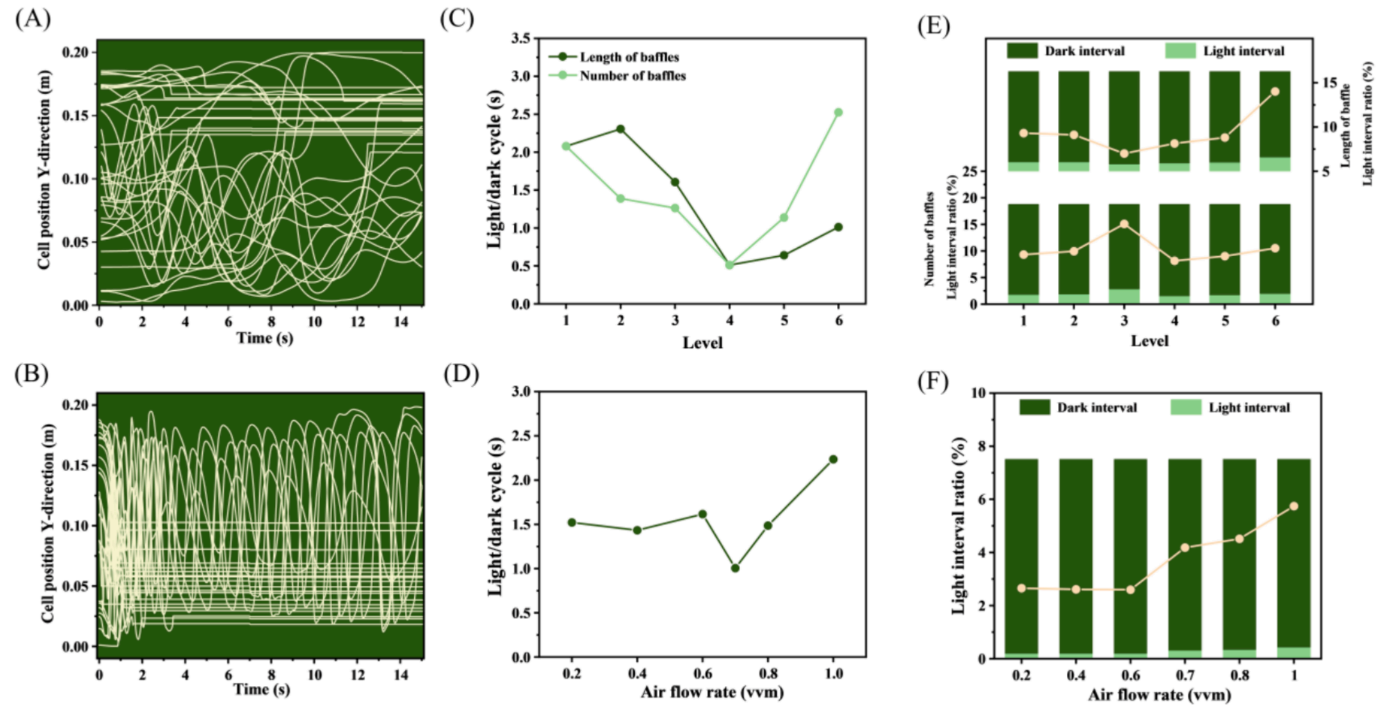

Fig. 5. Light parameters of the optimized reactor. (A) Trajectories of microalgae cells in the direction of light in the reactor in Reactor A and (B) Reactor B; (C) Light/Dark cycles for different baffle conditions; (D) Light/Dark cycles at different aeration rates; (E) Proportion of light zones for different baffle conditions; (F) Proportion of light zones under different aeration rates.

was actually prolonged, which may be due to the fact that the length was too short to form a local circulation. In terms of the number of baffles, the cycle time was inversely proportional to the number of baffles, which indicated that the stronger the vortex formed by the baffles was, the more beneficial was to the movement of microalgae cells, which was consistent with the previous research results (Fu et al., 2023). However, the gap was not much when the length of baffles increased from 6 cm to 8 cm, indicating that the length of baffles had little effect on the L/D cycle. When the aeration rate was between 0.2 and 0.6 vvm, the L/D cycle was shortened by 26.9 %, 31.1 % and 22.3 %, respectively, compared with that before optimization (Fig. 5D). The difference between them was very small, indicating that aeration rate less than 0.6 vvm had little effect on the cycling of microalgae cells. The L/D cycle was significantly reduced when the aeration rate was 0.7 vvm, which was only 51.6 % of that before optimization, and the best effect was achieved.

When the number of baffles was 14, the light was best, reaching 10.7 %. Fig. 5E shown that the number of baffles had little effect on the proportion of light zone. Baffles with a length of 4 cm result in the largest proportion of the time zone, which was related to its L/D cycle. This may be due to the fact that although the particle was able to stay in the light zone by the upward force, the cycling effect was insufficient. However, the proportion of time zones with baffle length of 6 cm and 8 cm was lower than that without baffles. This may be due to the relatively complete movement trajectory of microalgae in the vertical direction, because only a very small part of the reactor was defined as light zone (>0.175 m). The light time shown in Fig. 5F increased with the aeration rate, producing a large breakthrough at an aeration rate of 0.7 vvm.

3.3. Experimental verification

Based on the model data obtained by CFD calculations, the optimal reactor design scheme (RB) and the initial RA were selected for a control experiment to verify the simulation results. The number of baffles in RB was 14, the length was 6 cm, and the aeration rate was 0.7 vvm. The two reactors ran for about 400 h under the same conditions.

The removal of NH_4^+-N and NO_3^--N was shown in Fig. 6A and B. After culture, the NH_4^+-N in RA was reduced to less than 15 mg/L within 96 h, which met the discharge standard. The removal rates of NH_4^+-N in RA and RB reached 0.76 mg h^{-1} L^{-1} and 1.56 mg h^{-1} L^{-1}, respectively, and the optimized reactor NH_4^+-N rate increased by approximately 105.6 %. The removal of NH_4^+-N by microalgae was mainly via assimilation. The optimization of flow state in the reactor affected the photosynthetic activity of microalgae and thus promotes NH_4^+-N. The increase of NO_3^--N in RB was less than in RA (Fig. 6D), because the conversion of NH_4^+-N was not all due to assimilation as part of it was completed by nitrification (Liu et al., 2023). The assimilation of microalgae in RB was stronger, while nitrification was stronger in RA, which resulted in a higher NO_3^--N concentration (Liu et al., 2022). During the treatment, the concentration of NO_2^--N<2 mg/L, so it was not mentioned.

In addition, the change of biomass during the operation of the reactor also confirmed the optimization effect of RB (Fig. 6C). After 395 h of operation, the biomass in RA increased from 0.06 g/L to 3.01 g/L, and that in RB increased to 3.94 g/L, which was 1.3 times of that in RA. The growth trend of microalgae in each group was consistent with the biological S-type growth curve. The growth curve fitted the Logistic function well, and the fitting degree R^2 of the relationship between dry weight accumulation and time was higher than 0.99 (see Supplementary Material). Although the biomass was relatively similar at the later stage of operation, it was clear that the biomass in RB increased at a faster rate, which was shown in Fig. 6D. Through the analysis of the dynamic curve diagram of microalgae in the reactor, it shown that the point of the highest growth rate of biomass in RB was significantly higher than that in RA, indicating that the optimized reactor was more conducive to the growth and proliferation of microalgae. In addition, the biomass growth rate in RB was always higher than in RA at the early stage, which also verified the results obtained by the previous experiment to a certain extent.

4. Conclusion

In this study, the CFD method was employed to optimize the internal

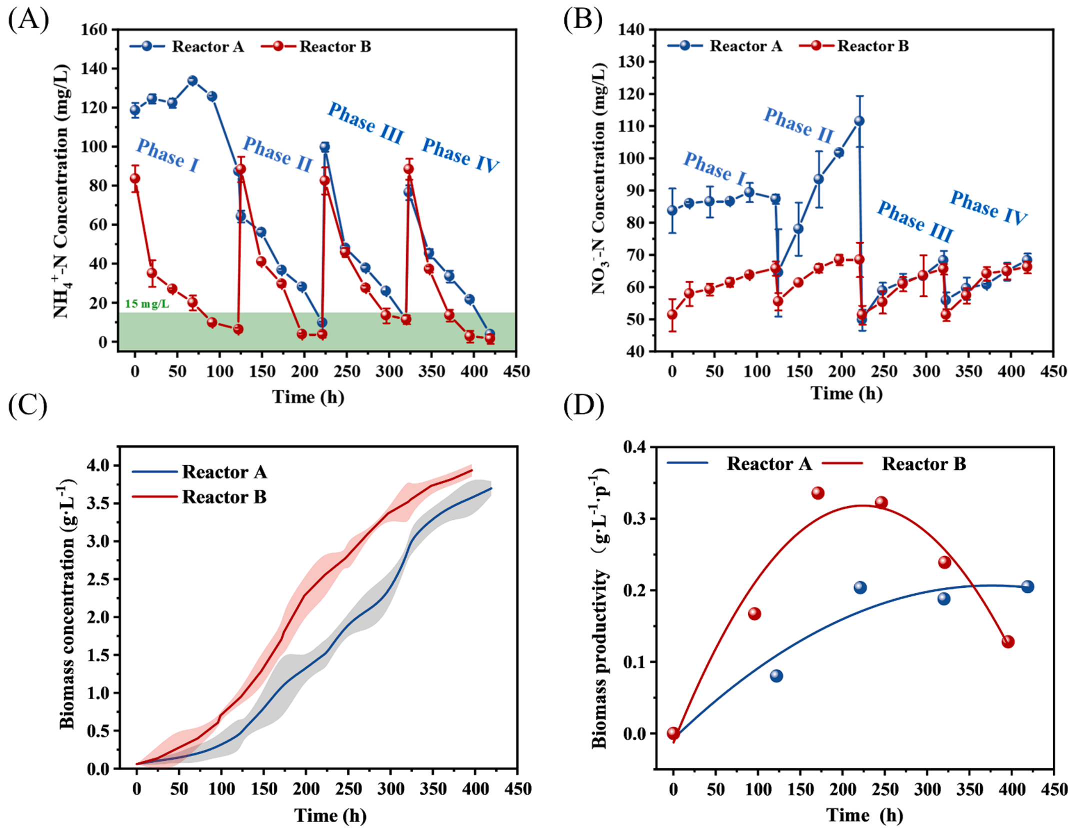

Fig. 6. Wastewater treatment effect and biomass growth in the Reactor A and Reactor B. (A) NH_4^+-N removal; (B) NO_3^--N production; (C) Biomass growth; (D) Biomass productivity. The error bars represent three replications (n = 2).

baffles in terms of their number, length, as well as the aeration rate, thereby ensuring the optimal growth of microalgae to their, fullest potential. The optimization of baffle and aeration rate improves liquid mixing, which improved the mixing degree of liquid in the reactor, reduced the dead zone area and accelerated the vertical movement speed. This study holds significant implications for advancing the practical implementation of microalgae wastewater treatment in engineering applications.

CRediT authorship contribution statement

Yuanqi Liu: Writing – review & editing, Writing – original draft. **Zhuochao Liu:** Writing – review & editing. **Zhensheng Xiong:** Methodology, Investigation, Funding acquisition, Conceptualization. **Yanni Geng:** Writing – review & editing, Visualization. **Dan Cui:** Writing – review & editing. **Spyros G. Pavlostathis:** Writing – review & editing. **Houxing Chen:** Project administration. **Qingchun Luo:** Project administration. **Genping Qiu:** Project administration. **Qiaohong Dong:** Project administration. **Liming Yang:** Writing – review & editing, Methodology, Investigation, Conceptualization. **Penghui Shao:** Validation. **Hui Shi:** Validation. **Xubiao Luo:** Methodology, Funding acquisition, Conceptualization. **Shenglian Luo:** Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2024.131293.

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